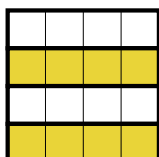
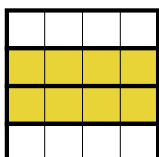
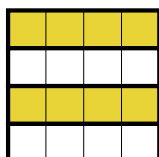
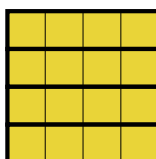


Polyomino Covers & Critical Sets, $n = 4$

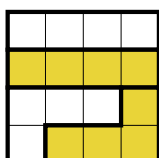
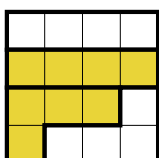
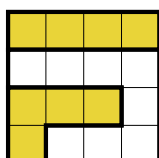
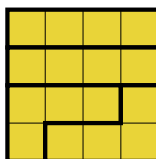
Colin Ramsay, perl v5.34.0

July 22, 2026

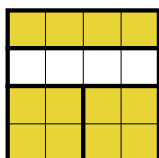
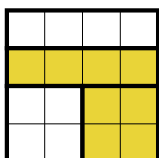
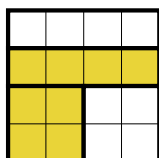
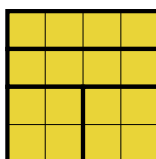
C4.1 4
AAAA BBBB CCCC DDDD
AC BC BD



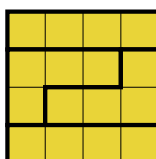
C4.2 1
AAAA BBBB CCCC DDDD
AC BC BD

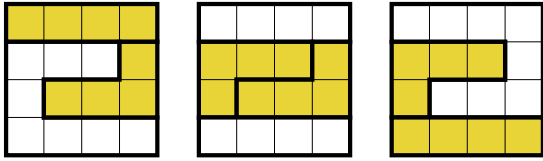


C4.3 2
AAAA BBBB CCCC DDDD
BC BD ACD

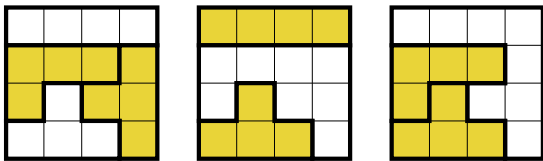
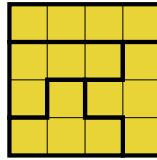


C4.4 2
AAAA BBBC BCCC DDDD
AC BC BD

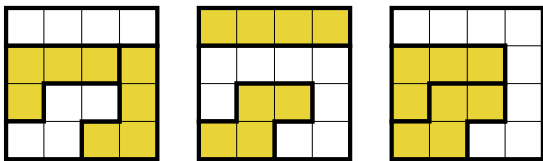
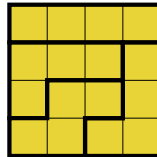




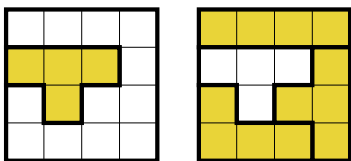
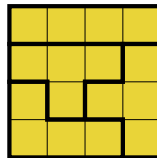
C4.5 1
AAAA BBBC BDCC DDDC
BC AD BD



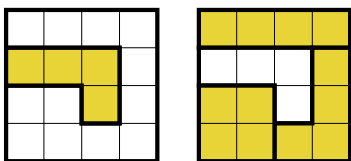
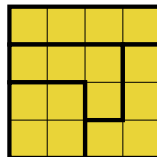
C4.6 1
AAAA BBBC BDDC DDCC
BC AD BD



C4.7 1
AAAA BBBC DBCC DDDC
B ACD



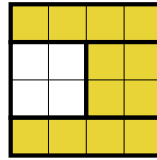
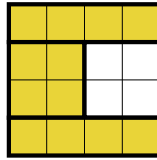
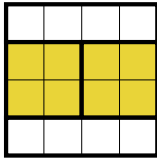
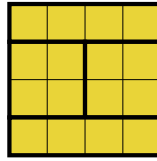
C4.8 1
AAAA BBBC DDBC DDCC
B ACD



C4.9 4

AAAA BBCC BBCC DDDD

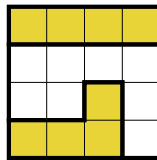
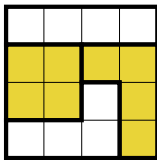
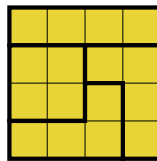
BC ABD ACD



C4.10 1

AAAA BBCC BBDC DDDC

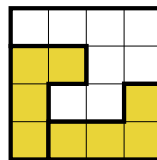
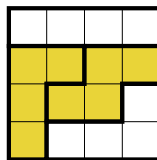
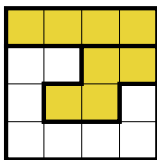
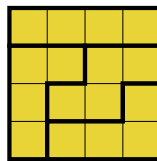
BC AD



C4.11 1

AAAA BBCC BCCD BDDD

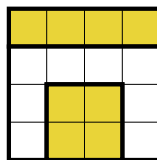
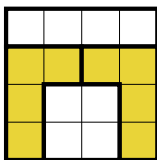
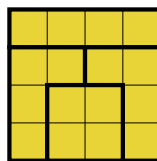
AC BC BD



C4.12 2

AAAA BBCC BDDC BDDC

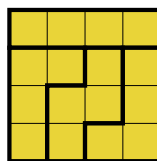
BC AD

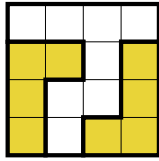
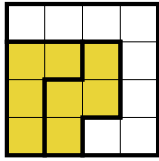
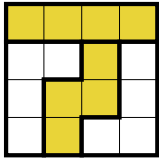


C4.13 1

AAAA BBCC BCCD BCDD

AC BC BD

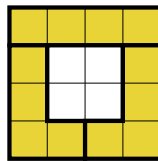
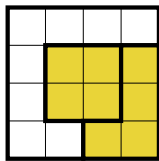
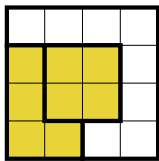
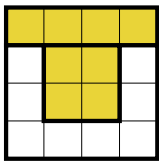
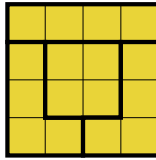




C4.14 2

AAAA BCCD BCCD BBDD

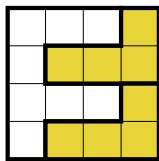
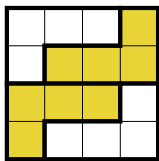
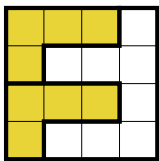
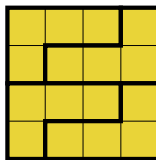
AC BC CD ABD



C4.15 2

AAAB ABBB CCCD CDDD

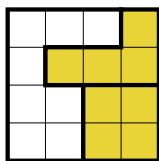
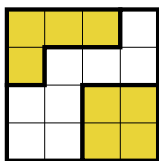
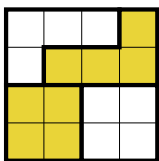
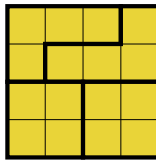
AC BC BD



C4.16 1

AAAB ABBB CCDD CCDD

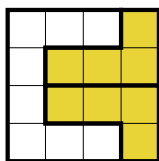
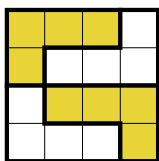
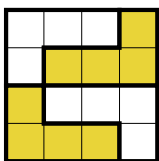
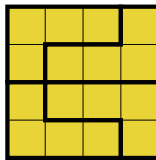
BC AD BD



C4.17 2

AAAB ABBB CDDD CCCD

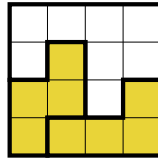
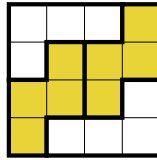
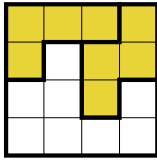
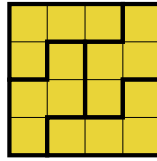
BC AD BD



C4.18 2

AAAB ACBB CCBD CDDD

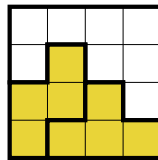
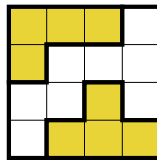
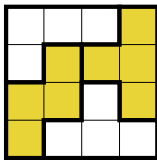
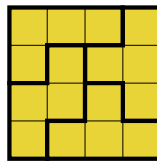
AB BC CD



C4.19 1

AAAB ACBB CCDB CDDD

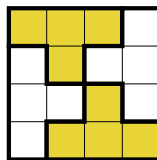
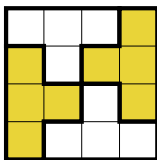
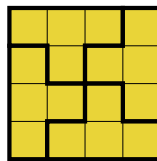
BC AD CD



C4.20 4

AAAB CABB CCDB CDDD

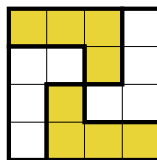
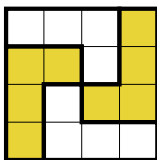
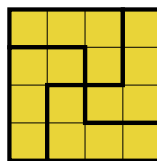
BC AD



C4.21 4

AAAB CCAB CDBB CDDD

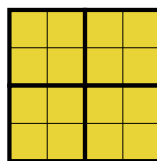
BC AD

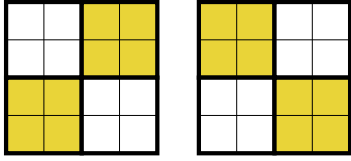


C4.22 8

AABB AABB CCDD CCDD

BC AD





Order = 4, SCS = 1, LCS = 3

Free: Covers = 22, Total CS = 60, CS per Cover = 2 ... 4

Fixed: Covers = 117, Total CS = 322